

INDEFINITE INTEGRATION FORMULAE:

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$$

$$2. \int dx = x + c$$

$$3. \int \frac{1}{x} dx = \log|x| + c$$

$$4. \int a^x dx = \frac{a^x}{\log a} + c$$

$$5. \int e^x dx = e^x + c$$

$$6. \int \sin x dx = -\cos x + c$$

$$7. \int \cos x dx = \sin x + c$$

$$8. \int \sec^2 x dx = \tan x + c$$

$$9. \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$10. \int \sec x \tan x dx = \sec x + c$$

$$11. \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$$

$$12. \int \tan x dx = -\log|\cos x| + c \text{ or } \log|\sec x| + c$$

$$13. \int \cot x dx = \log|\sin x| + c$$

$$14. \int \sec x dx = \log|\sec x + \tan x| + c \text{ or } \log\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + c$$

$$15. \int \operatorname{cosec} x dx = \log|\operatorname{cosec} x - \cot x| + c \text{ or } \log\left|\tan\frac{x}{2}\right| + c$$

$$16. \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$$

$$17. \int -\frac{1}{\sqrt{1-x^2}} dx = \cos^{-1} x + c$$

$$18. \int \frac{1}{1+x^2} dx = \tan^{-1} x + c$$

$$19. \int -\frac{1}{1+x^2} dx = \cot^{-1} x + c$$

$$20. \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + c$$

$$21. \int -\frac{1}{x\sqrt{x^2-1}} dx = \cos ec^{-1} x + c$$

ALGEBRA OF INTEGRATION

$$1. \int [f_1(x) \pm f_2(x)] dx = \int f_1(x) dx \pm \int f_2(x) dx$$

$$2. \int k \cdot f(x) dx = k \int f(x) dx$$

INTEGRATION BY PARTS:

Integration by parts is used in integrating functions of the type $f(x) \cdot g(x)$ or $u \cdot v$ as follows.

$$\int (I^{st} \text{ function} \times II^{nd} \text{ function}) dx = I^{st} \text{ function} \int (II^{nd} \text{ function}) dx - \int \left(\frac{d}{dx} (I^{st} \text{ function}) \times \int (II^{nd} \text{ function}) dx \right) dx$$

Where the Ist and IInd functions are decided in the order of LIATE;

L: Logarithmic function

I: Inverse trigonometric function

A: Algebraic functions

T: Trigonometric functions

E: Exponential Functions